

Bird Species Detection Using Deep Learning

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Abstract

This project focuses on automated bird species recognition using the CUB-200-2011 dataset and Vision Transformer (ViT-B/16). The work involves a pipeline consisting of dataset analysis, manual cleaning, augmentation, fine-tuning, and performance evaluation. Extensive experimentation demonstrated that dataset quality plays a crucial role in fine-grained classification. After implementing targeted cleaning steps, enhancing under-represented classes, and employing advanced augmentation techniques, the final trained model achieved an accuracy of **87.12%** on the test set.

A detailed misclassification analysis shows that **70.2% of errors occur within species belonging to the same bird family**, highlighting the complexity of subtle inter-class differences. Potential improvements include hierarchical classification, part-based attention models, and expansion of the dataset using verified external sources.

1 Introduction

Bird species identification is a crucial task in biodiversity conservation, ecological monitoring, and wildlife research. Traditionally, ornithologists rely on visual cues such as plumage patterns, beak shape, body proportions, and habitat characteristics. However, manual identification is slow, prone to human error, and often inaccessible to non-experts.

The rapid growth of machine learning and computer vision has made automated species recognition a promising solution. Deep learning architectures, particularly transformers, have shown remarkable performance in fine-grained recognition tasks where classes differ subtly.

This project applies Vision Transformer (ViT-B/16) to classify images from the CUB-200-2011 dataset, one of the most widely used benchmarks for fine-grained bird classification. The objective is to design a robust, scalable model that can analyze visual information and identify species with high accuracy.

2 Motivation

Several factors motivated the development of this project:

Challenges in Manual Identification

- Fine-grained species often differ by extremely subtle features such as eye rings, wing bars, or feather shades.
- Environmental variations like lighting, background clutter, occlusion, and camera angles make identification difficult.
- Availability of expert ornithologists is limited, making large-scale monitoring impractical.

Benefits of Automation

- Enables large-scale ecological studies and long-term biodiversity tracking.
- Assists non-experts (birdwatchers, students, citizen scientists) in species recognition.
- Supports conservation efforts by identifying endangered or invasive species quickly.

The combination of scientific importance and technological feasibility makes automated bird classification an impactful problem to solve.

3 Literature Review

3.1 Bird Species Categorization Using Pose-Normalized Deep CNN (Branson et al., 2014)

- Introduces pose-normalized CNN where bird images are aligned using part locations (beak, wings, tail).
- Reduces viewpoint and pose variation for improved fine-grained recognition.
- Achieves **75.7%** accuracy on the CUB-200-2011 dataset.

3.2 VGG Baselines for Fine-Grained Recognition (Simonyan & Zisserman, 2014)

- VGG-16 and VGG-19 act as strong feature extractors for fine-grained visual classification.
- Increased depth + small convolution filters improve hierarchical feature learning.
- Achieved accuracies:
 - **70%** (VGG-16)
 - **72%** (VGG-19)

3.3 Recurrent Attention CNN (RA-CNN) (Fu et al., 2017)

- Introduces a multi-stage attention mechanism that zooms progressively into discriminative image regions.

- Learns important local features without requiring part annotations.
- Achieves **85.3%** accuracy on CUB-200-2011.

3.4 Multi-Attention CNN (MA-CNN) (Zheng et al., 2017)

- Automatically discovers multiple informative regions using learned attention maps.
- Jointly optimizes part localization and fine-grained classification.
- Achieves **86.5%** accuracy on CUB-200-2011.

3.5 Transformer-Based FGVC (TransFGVC) (Shen et al., 2022)

- Uses Swin Transformer with LSTM refinement to enhance contextual feature understanding.
- Suppresses background noise and focuses on finer bird features.
- Achieves **92.58%** accuracy, outperforming most CNN-based methods.

3.6 SupCon-ViT: Supervised Contrastive Learning (Lu et al., 2023)

- Applies supervised contrastive loss to cluster same-class embeddings more tightly.
- Highly effective for ultra-fine-grained species with minimal inter-class differences.
- Significantly improves ViT performance compared to standard training.

3.7 Dataset Flaws & Label Noise in CUB-200-2011 (Van Horn et al., 2015)

- Reports that **4%** of CUB-200-2011 images are mislabeled or ambiguous.
- Label noise limits maximum achievable accuracy for fine-grained classification.
- Flickr retrieval quality for FGVC datasets found to be only **84%** correct (16% mislabeled).

4 Dataset Details

CUB-200-2011 Overview

- Total Images: **11,788**
- Classes: **200 bird species**
- Training Images: **5,994**
- Testing Images: **5,794**
- Metadata: Class labels, image paths, bounding boxes, part annotations

The dataset is highly imbalanced, with some classes containing over 60 images and others barely reaching 30. To improve training stability, a custom **80:10:10 stratified split** was created, ensuring that minority classes were better represented in validation and testing.

Additional cleaning was required to remove mislabeled and corrupted images, based on visual inspection and feature-space anomaly detection.

5 Methodology

5.1 Dataset Preparation

Metadata files were merged to associate each image with its bounding box and class label. A stratified split was applied to preserve proportional representation.

5.2 Data Cleaning

More than 60 mislabeled or low-quality images were identified using:

- t-SNE and PCA outlier detection
- Manual verification
- Checking for bounding box inconsistencies

5.3 Dataset Extension

To reduce class imbalance, additional images from publicly available datasets were manually added for weak classes after verifying species authenticity.

5.4 Data Augmentation

Key augmentations included:

- RandomResizedCrop for spatial diversity
- Horizontal and slight vertical flips
- Small-angle rotations to simulate pose variations
- Color jittering for lighting normalization
- Normalization based on ImageNet mean and std

5.5 Model Selection

ViT-B/16 was selected due to:

- Strong performance in fine-grained recognition
- Global receptive field due to self-attention
- Suitability for clean datasets with high inter-class similarity

5.6 Training Strategy

- **Optimizer:** AdamW with weight decay
- **Scheduler:** Cosine annealing with warm restarts
- **Batch Size:** 32
- **Epochs:** 50 (early stopping triggered earlier)
- **Mixed Precision:** Enabled via FP16/AMP
- **Gradient Clipping:** Prevented exploding gradients

5.7 Evaluation

Metrics used:

- Overall accuracy
- Per-class accuracy
- Precision, recall, F1-score
- Confusion matrix
- Misclassification clustering

6 Progress Till Now

Milestone Summary

- **ResNet18 baseline:** 55%
- **ViT-B/16 with minimal augmentation:** 84%
- **Stronger augmentation:** 86%
- **Custom dataset split:** 88%
- **Added external images:** 89.4%
- **After cleaning mislabeled images:** 87.12%

Each improvement step highlighted the importance of dataset quality. Interestingly, cleaning mislabeled images reduced overall accuracy slightly due to reduced dataset size but created a *more reliable* evaluation environment.

7 Results

7.1 Final Accuracy

87.12%

7.2 Misclassification Analysis

- **70.2% of incorrect predictions occur within the same bird family.**
- Indicates difficulty in distinguishing species with minimal visual differences.
- Confusion was highest among:
 - Warblers
 - Sparrows
 - Thrushes
- Models struggled with:
 - Poor lighting
 - Occlusions
 - Juvenile or non-breeding plumage

This suggests that incorporating part-based attention (e.g., focusing on head and wings) may improve performance.

7.3 Accuracy Progression

To better understand the improvement across different stages of model development, Table 7.1 summarizes the accuracy achieved at each milestone.

Table 7.1: Accuracy Achieved at Each Stage of Model Development

Stage	Accuracy (%)
ResNet18 Baseline	55.00
ViT-B/16 with Minimal Augmentation	84.00
ViT-B/16 with Stronger Augmentation	86.00
Custom Stratified Dataset Split	88.00
Added External Images for Weak Classes	89.40
After Removing Mislabeled Images	87.12
After Removing Mislabeled Images and keeping 50:50 Split	87.15

7.4 Visual Results

7.4.1 ViT-B/16 Architecture

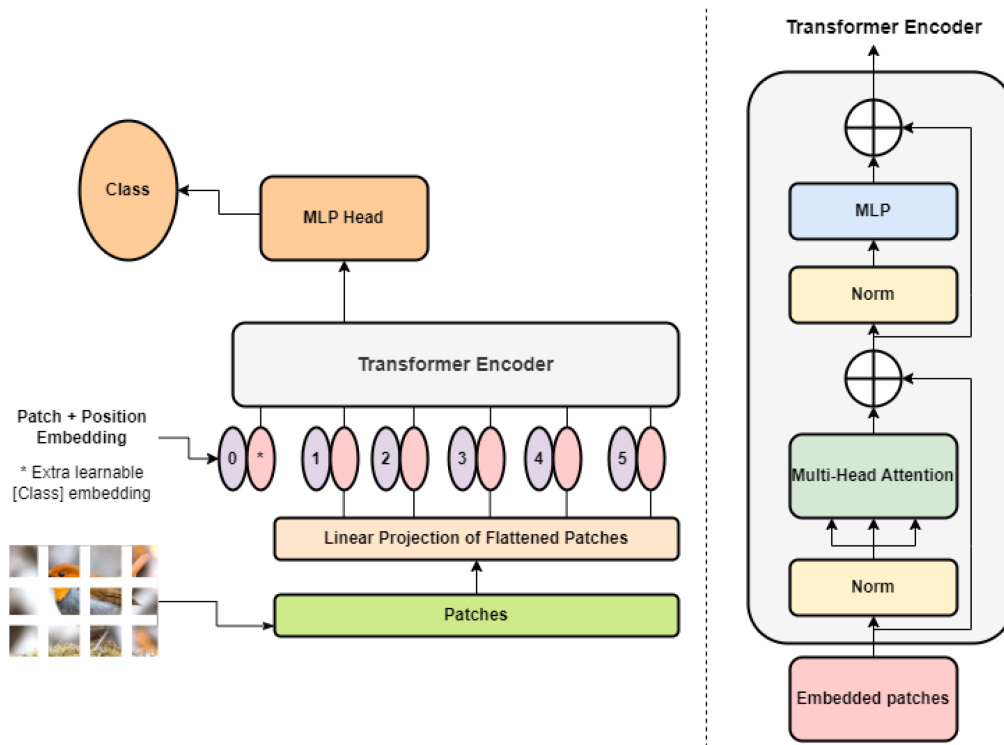


Figure 7.1: ViT-B/16 Architecture

Description:

- The diagram illustrates the overall architecture of the Vision Transformer (ViT) used for image classification.
- The input image is first divided into fixed-size **patches**, which are then flattened and passed through a **linear projection layer**.
- A learnable **[CLS] token** is prepended to the sequence of patch embeddings; this token is responsible for accumulating global image information for final classification.
- **Positional embeddings** are added to each patch embedding to retain spatial information lost during patch extraction.
- The sequence of embedded patches (including the CLS token) is fed into a stack of **Transformer Encoder** blocks.
- Each encoder block contains:

- **Layer Normalization**
 - **Multi-Head Self-Attention** mechanism to allow global contextual understanding across image patches.
 - A second **Normalization** layer
 - A **Feed-Forward MLP** sub-layer
 - **Residual connections** around both the attention and MLP modules
- After several transformer layers, the output corresponding to the **CLS token** is extracted.
 - This final CLS embedding is passed through an **MLP classification head** to produce the predicted class label.
 - The architecture demonstrates how ViT leverages self-attention to model long-range dependencies across image regions without using convolution.

7.4.2 Accuracy vs Epoch Graph

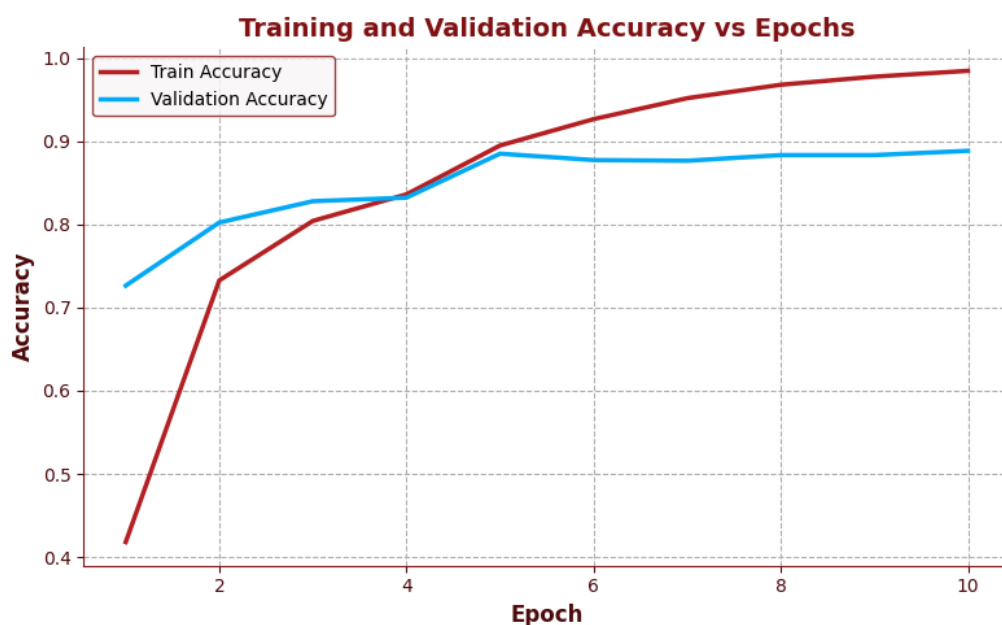


Figure 7.2: Training and Validation Accuracy vs Epochs

Description:

- The graph plots the **training accuracy** and **validation accuracy** across multiple epochs.
- Training accuracy shows a **consistent upward trend**, indicating steady learning throughout the training process.

- Validation accuracy increases initially and then **stabilizes**, reflecting the model's generalization capability.
- The gap between the two curves indicates the level of **overfitting or underfitting**.
- The curves converge toward later epochs, suggesting that the model achieves **stable and reliable** performance.
- Overall, the graph demonstrates effective convergence of the ViT-B/16 model on the dataset.

7.4.3 Augmentation Types



Figure 7.3: Overview of Data Augmentation Techniques Used

Description:

- The figure showcases the various data augmentation techniques applied during training.
- Augmentations include RandomResizedCrop, horizontal flipping, minor rotations, vertical shift, and color jitter.
- These transformations increase dataset diversity and reduce overfitting.
- Visual examples illustrate how augmentations simulate real-world variations in lighting, pose, and background.
- The use of multiple augmentation types helps the model generalize better to unseen bird images.

7.4.4 Confusion Matrix

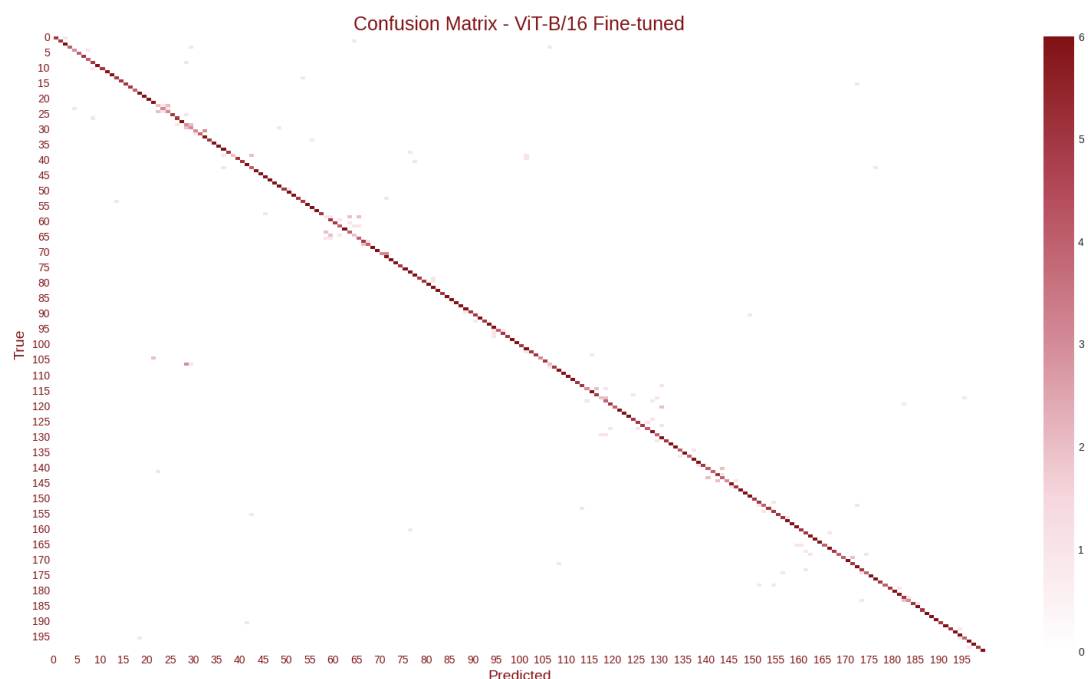


Figure 7.4: Confusion Matrix for All 200 Bird Classes

Description:

- The confusion matrix visualizes the model's predictions across all 200 bird species.
- Diagonal cells represent **correct classifications**, where predicted labels match true labels.
- Off-diagonal cells indicate **misclassifications**, showing which species the model tends to confuse.
- Darker diagonal regions indicate strong performance on most classes.
- Misclassifications are more frequent among visually similar species, especially within the same bird family.
- The matrix highlights patterns of systematic confusion, useful for guiding dataset cleaning or model improvements.

7.4.5 Family Distribution: Same Family vs Different Family Misclassifications

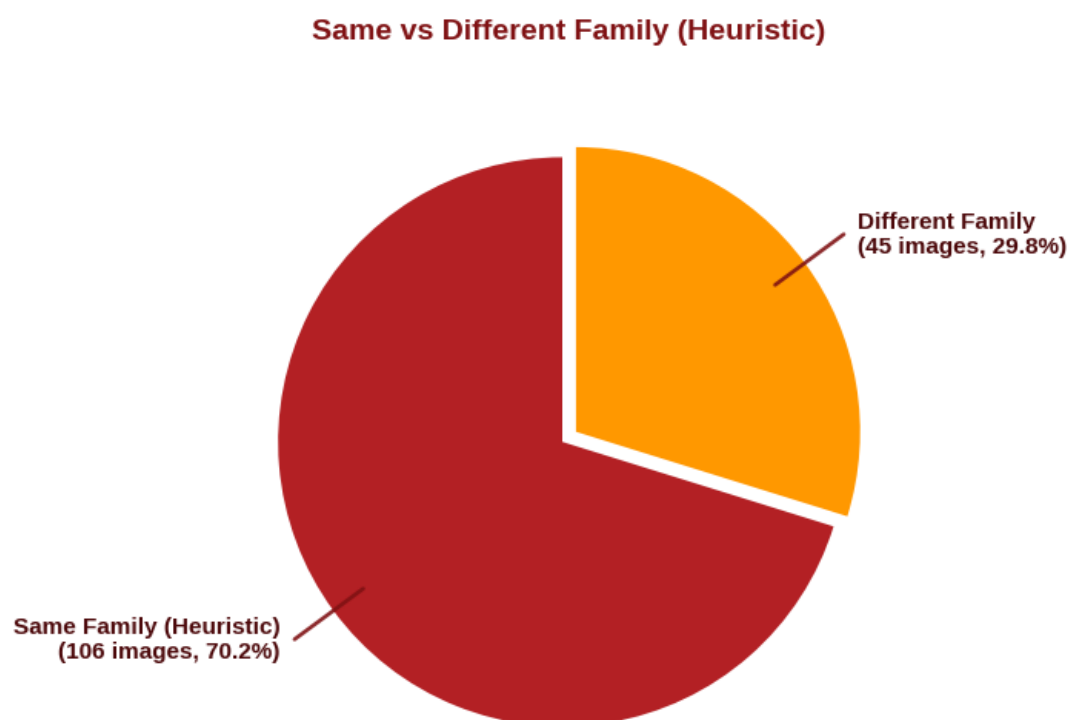


Figure 7.5: Misclassification Distribution: Same Family vs Different Family

Description:

- The graph compares misclassifications occurring **within the same family** and **across different families**.
- A significantly higher percentage of errors occur within the same family.
- This highlights the challenge of **fine-grained classification**, where species share subtle visual similarities.
- Very few misclassifications occur across unrelated families, indicating strong inter-class separation.
- The distribution emphasizes the need for **part-based attention** or **hierarchical classification**.

7.4.6 Training, Validation, and Test Loss Graph



Figure 7.6: Training, Validation, and Test Loss over Epochs

Description:

- The graph shows **training, validation, and test loss** decreasing over epochs.
- Training loss consistently decreases, indicating that the model is learning effectively.
- Validation and test loss follow a similar trend but with **minor fluctuations**, reflecting dataset complexity.
- The curves converge gradually, showing **stable generalization** across unseen data.
- No large divergence is observed, indicating the absence of severe overfitting.

7.4.7 Sample Misclassified Images

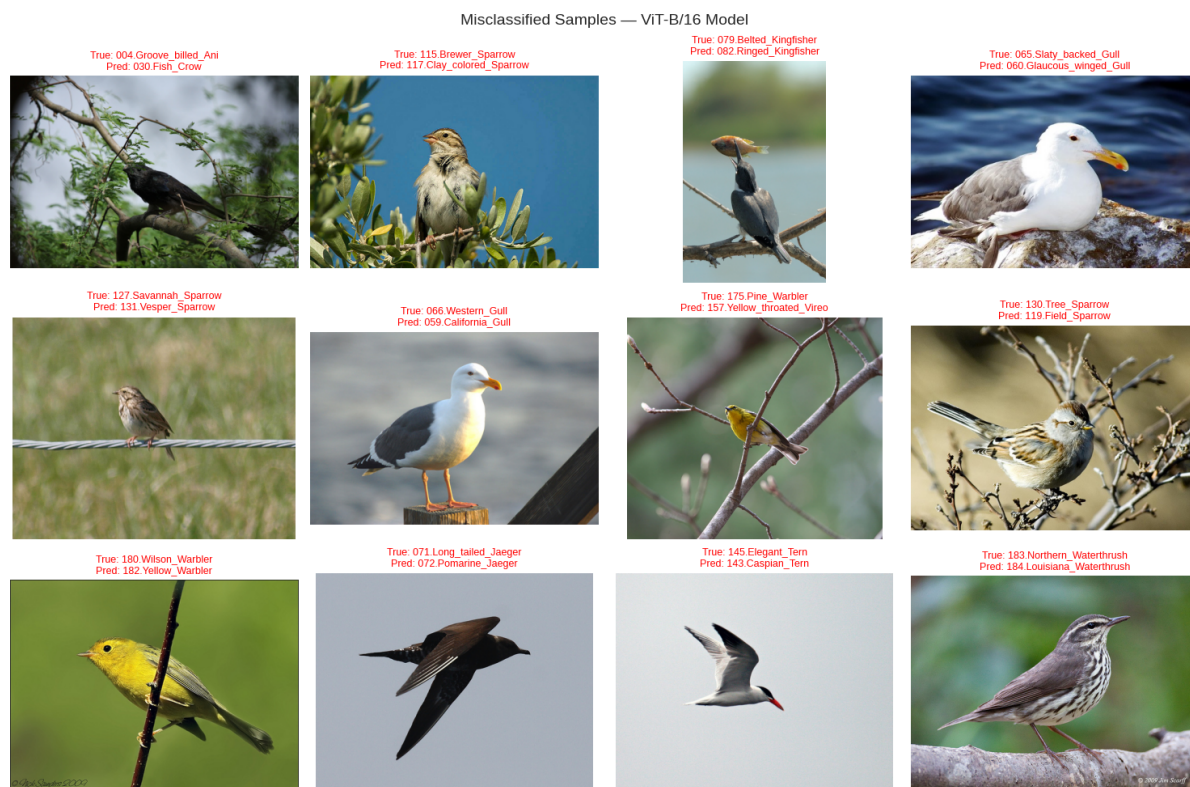


Figure 7.7: Examples of Misclassified Images and Model Predictions

Description:

- The figure shows sample images where the model predicted incorrect species labels.
- Each example includes the **true label** and the model's **predicted label**.
- Many misclassifications occur among species with similar plumage, color tones, or body shapes.
- Several errors are linked to challenging conditions such as poor lighting, occlusion, and complex backgrounds.
- These visual results help identify limitations in the model's fine-grained discrimination ability.

7.4.8 Train-Test Split Distribution

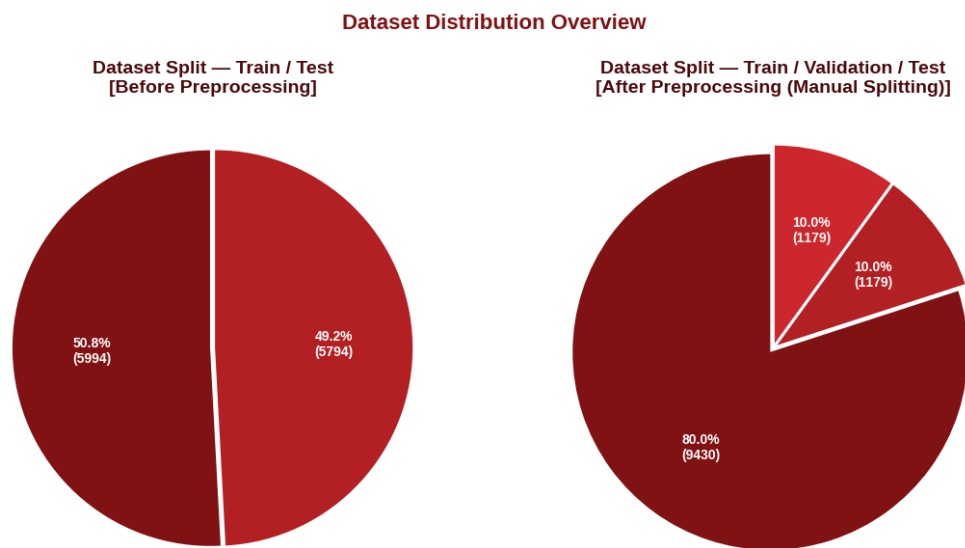


Figure 7.8: Distribution of Training, Validation, and Test Samples

Description:

- The graph compares the distribution of training, validation, and test samples before and after dataset cleaning.
- It visualizes the class-wise sample counts to highlight dataset imbalance.
- After stratified splitting, each class maintains proportional representation across train, validation, and test sets.
- The post-cleaning distribution shows removal of mislabeled and low-quality images.
- The visualization confirms improved dataset structure and more reliable evaluation after cleaning.

7.4.9 Workflow Flowchart

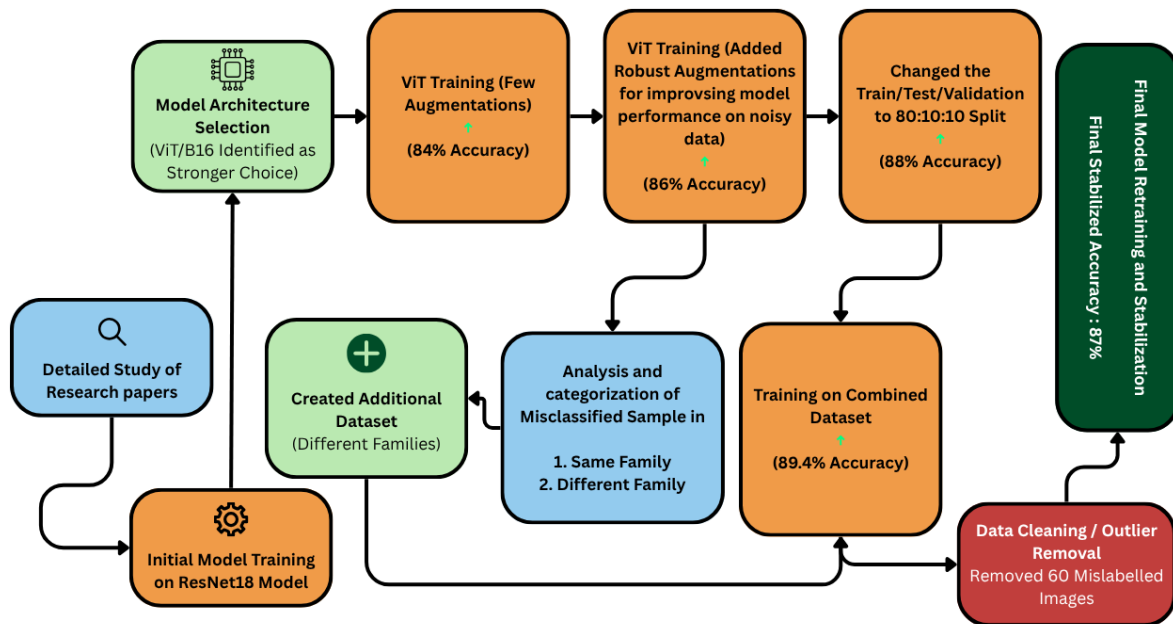


Figure 7.9: Complete Workflow of the Bird Species Detection Pipeline

Description:

- The flowchart presents the complete pipeline of the Bird Species Detection project.
- It outlines each major stage: dataset loading, cleaning, augmentation, splitting, model training, and evaluation.
- Arrows indicate the sequential workflow from raw images to final predictions.
- The diagram highlights dependencies between processing steps in a clear, structured manner.
- The flowchart provides a high-level understanding of how data flows through the system.

8 Expected Future Work

8.1 Dataset Improvements

- Manual expert verification for ambiguous samples.
- Incorporating additional high-quality images from iNaturalist, GBIF, or Macaulay Library.
- Removing images with extreme occlusions or background clutter.

8.2 Model Generalization

- Cross-evaluating model performance on external datasets.
- Exploring large-scale transformers like ViT-L/16 or Swin-L.

8.3 Fine-Grained Improvement

- Applying hierarchical classification (family $\rightarrow$ genus $\rightarrow$ species).
- Adding part-based feature extraction modules.
- Incorporating bird pose estimation (e.g., wing and tail keypoints).

9 Conclusion

The project demonstrates that Vision Transformers are highly effective for fine-grained bird species classification. Despite challenges related to dataset imbalance, noise, and inter-class similarity, the final model reached an accuracy of 87.12%. The results show that dataset quality and augmentation strategies significantly influence performance.

Future work will focus on hierarchical classification, dataset expansion, and advanced attention mechanisms to further improve accuracy and real-world robustness.

References

- Branson et al. (2014). *Bird Species Categorization Using Pose-Normalized Deep Convolutional Nets*
- Fu et al. (2017). *Look Closer to See Better: Recurrent Attention for Fine-Grained Categorization.*
- Simonyan & Zisserman (2014). *Deep Convolutional Networks for Fine-Grained Recognition (VGG Baselines).*
- Zheng et al. (2017). *Learning Multi-Attention CNN (MA-CNN) for Fine-Grained Recognition.*
- Lu et al. (2023). *SupCon-ViT: Supervised Contrastive Learning for Ultra FGVC.*